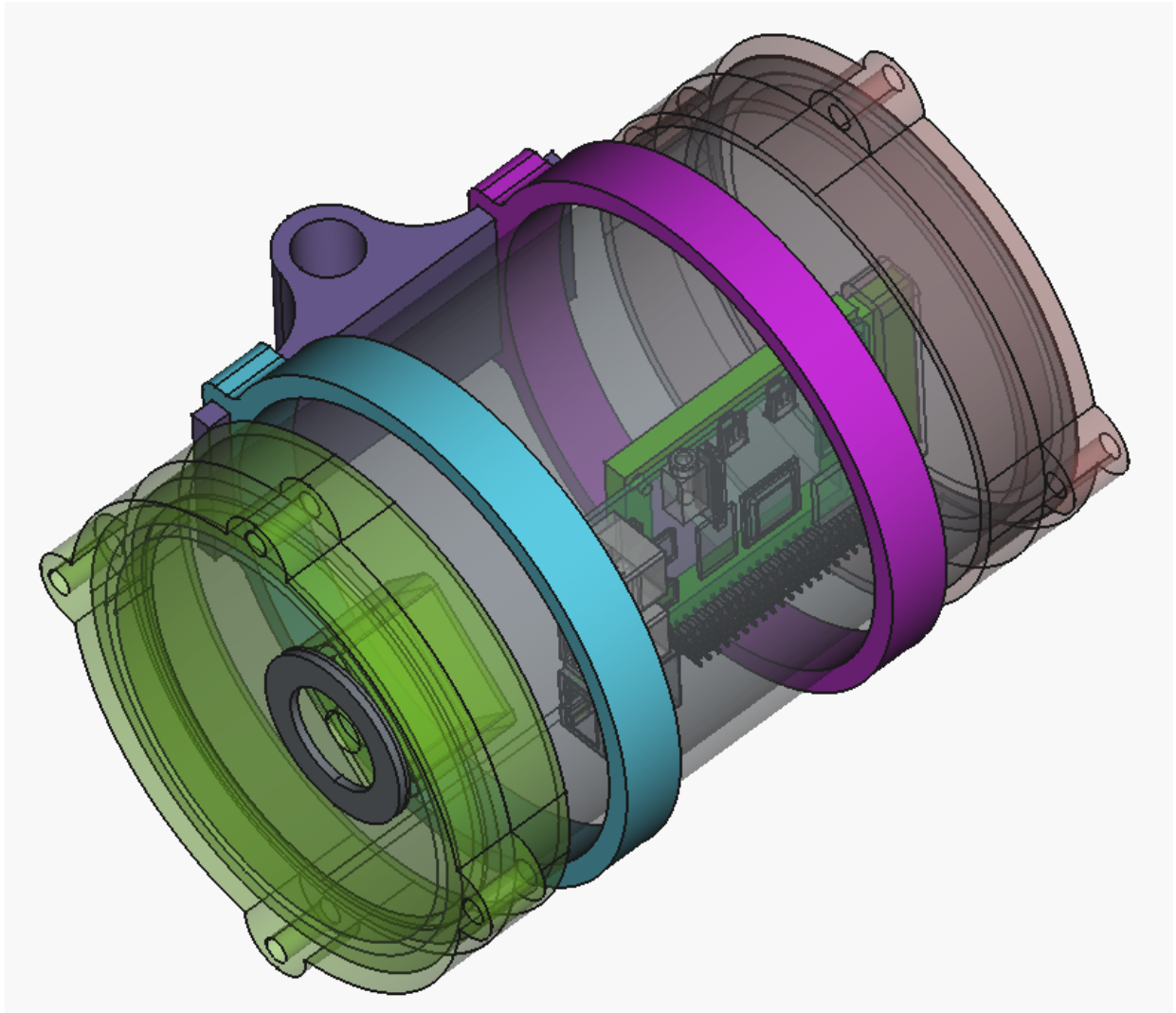




BIRDPIPE 1.0 ASSEMBLY INSTRUCTIONS

This document includes information about the first version of the BirdPipe assembly and its material costs. Document was written on **16 August 2026**; prices and component availability may vary in the future.

BirdPipe is an autonomous, Edge AI-capable bird and bat sound recorder based on a Raspberry Pi and USB audio devices. This guide explains how to assemble the first BirdPipe v. 1.0, choose compatible components, protect the microphone and electronics, and prepare the device for outdoor acoustic monitoring.

REQUIREMENTS

The components and prices below describe the prototype configuration used in the test device. They are not minimum requirements for every BirdPipe build. Lower-cost alternatives may be suitable depending on the microphone type, required AI-model memory consumption, power source, enclosure method, and field conditions or need of ultrasound recording.

Component	Used component	Price
Raspberry Pi	Raspberry Pi 4 Model B 8 GB	180,00 €
USB microphone	Dodotronic Ultramic 384K EVO	280,00 €
Power source	Wave Ultrapower 30.000 mAh	39,90 €
Optional: GNSS module	Waveshare SX1262	24,99 €
Optional: LTE	Brovi (Huawei) E3372-325 LTE/4G/3G USB	39,99 €
Optional: Acoustic vent	Gore Acoustic Vent AVP series	5,00 €
3D filament	Generic PETG (white)	5,00 €
110 mm PVC pipe	110 mm waste pipe (20 cm)	2,00 €
USB cables	Generic USB cables	20,00 €
Total cost per device	-	596,88 €

SHELL STRUCTURE



The shell consists of the following four components:

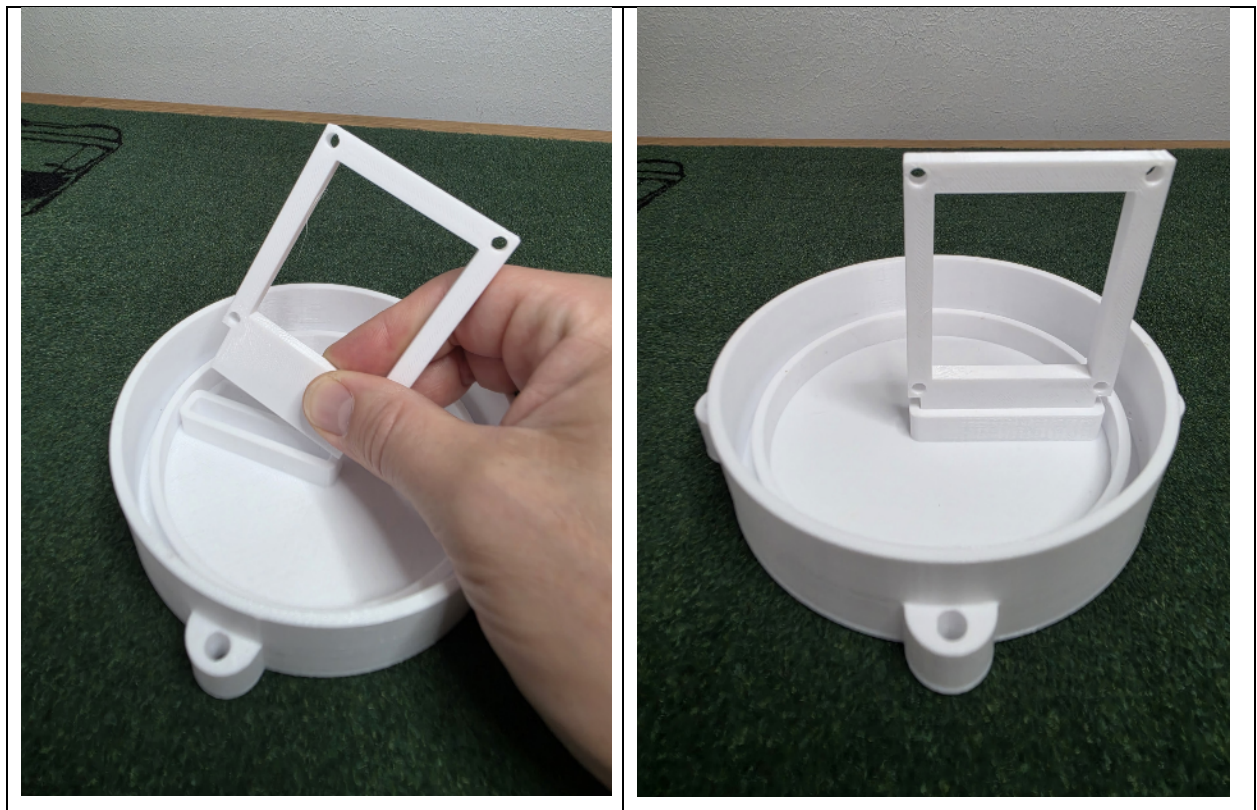
- 110 mm waste pipe
- `printable_parts_and_instructions/BirdPipe-110mmEndForDodotronic.stl`
- `printable_parts_and_instructions/BirdPipe-110mmEndForRaspHolder.stl`
- `printable_parts_and_instructions/BirdPipe-RaspberryPiHolder.stl`

The cost of a 110 mm PVC waste pipe in Finland is approximately 18,00 € for a 1 m section. Only 20 cm of pipe is required for a single BirdPipe unit, making the housing material cost very low. The pipe can be cut easily using a handsaw, jigsaw, or a dedicated pipe-cutting tool. When cutting or handling the pipe, always use appropriate protective equipment, including safety glasses and protective gloves. Remove and round any sharp edges with sandpaper after cutting to prevent injury and ensure a proper fit of the assembled components.

The required parts can be printed using the STL files provided above with a 3D printer capable of printing PETG or ABS materials. We recommend using white filament to reduce heat absorption from direct sunlight. Alternatively, standard pipe end caps and sealing components available from hardware stores can be used instead of the 3D-printed parts. For the BirdPipe prototype assembly described in this document, all parts were printed using a Bambu Lab X1 Carbon 3D printer.

As an alternative to 3D printing, the enclosure can be assembled using standard off-the-shelf pipe end caps and fittings. This removes the need for a 3D printer and allows the device to be built from commercially available plumbing components. Because pipe dimensions and end cap availability vary between countries and manufacturers, verify that the selected end cap fits securely onto the cut pipe before purchasing materials. A proper fit is important for mechanical stability and weather protection.

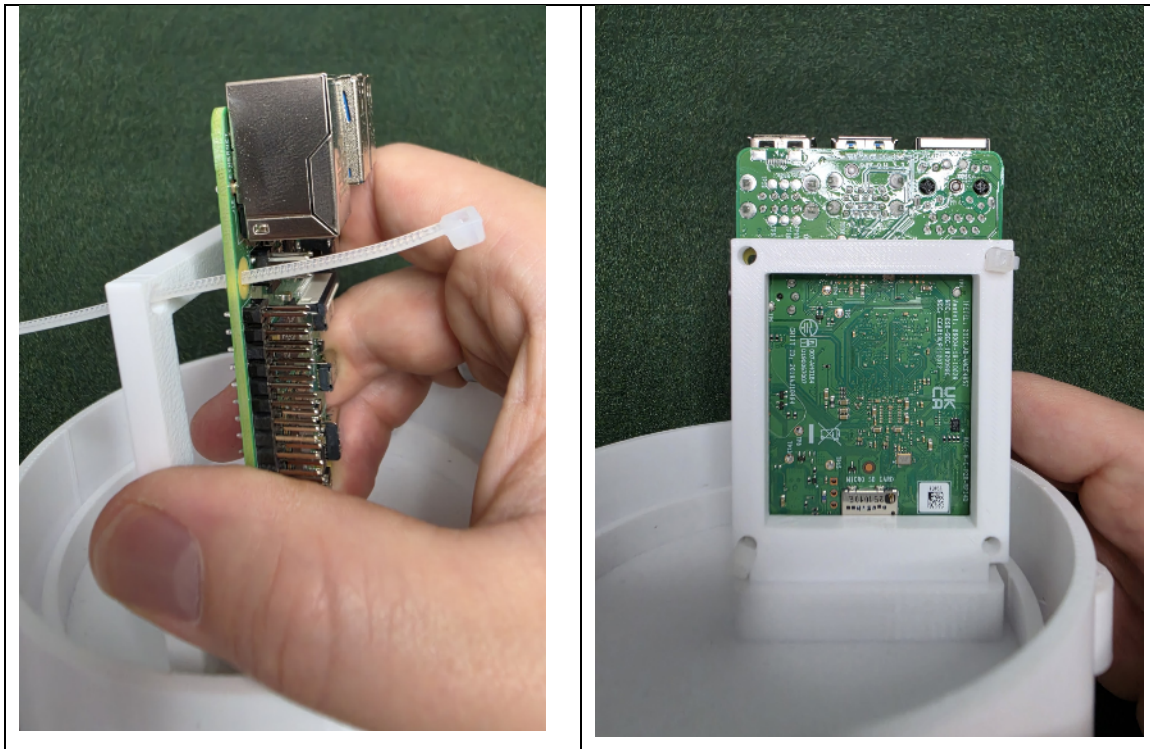
The end cap and holder components are designed for a press-fit assembly, allowing easy installation and removal. Under normal circumstances, no glue or adhesive is required.



RASPBERRY PI

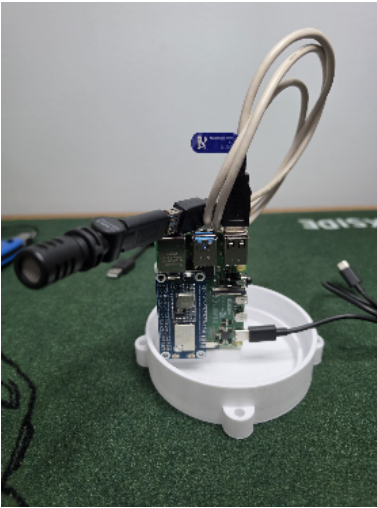
Mount the Raspberry Pi to the BirdPipe-RaspberryPiHolder using small zip ties passed through the mounting holes of the Raspberry Pi circuit board. In most cases, fastening two opposite corners provides adequate support and keeps the board securely positioned inside the enclosure. Ensure that the zip ties do not interfere with connectors, cables, or electronic components.

When assembling, ensure that the USB ports face upwards as shown in the following picture. Insert the SD card before tightening the zip ties.



ADDING MICROPHONE AND CABLES

In our device, we used the Dodotronic Ultramic 384K EVO, an ultrasonic microphone capable of recording frequencies beyond the range of human hearing. The microphone provides excellent audio quality but is relatively expensive, costing approximately 280,00 €. However, the BirdPipe system is based on Linux and is not limited to these specific microphone models. Most USB microphones compatible with Raspberry Pi and Raspberry Pi OS can also be used for audio recording, allowing users to choose a more affordable option depending on their requirements.

		
Sound Blaster audio card and analog microphone	Unnamed USB microphone	Dodotronic

The structure `printable_parts_and_instructions/BirdPipe-110mmEndForDodotronic.stl` holds the Dodotronic microphone securely, but with a drill, hot glue, and some modification skills, it is easy to adapt the structure for another microphone variant to significantly reduce costs.

Another cable is used for USB power. In our prototype, we used a 30,000 mAh power bank. Alternatively, the device can be powered from a wall outlet using a USB power adapter, or from a solar-powered system using an appropriate USB or 12 V power converter. It is important to understand the safety considerations associated with batteries, solar panels, and electrical installations. Always follow the manufacturer's instructions and applicable local regulations. If you are unsure about any aspect of the installation, consult a qualified local specialist for guidance and safety information.

MICROPHONE PROTECTION

The open microphone hole is a vulnerable part of the enclosure. Care should be taken to prevent water, dust, insects, and other debris from entering the device through the microphone opening. The BirdPipe enclosure is not fully waterproof, and additional protection may be required depending on the installation environment. Your need for protection may vary depending on the research environment.

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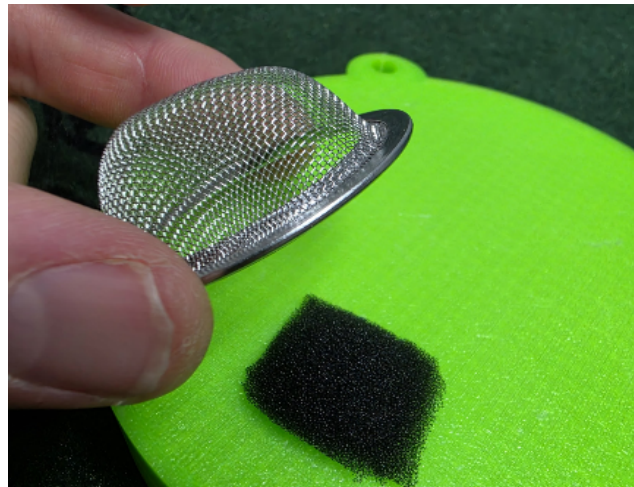
The Gore Acoustic Vent AVP series membrane (1 cm diameter) helps protect the device from rain and water splashes. The membrane corners can also be secured using general-purpose super glue. A small amount of foam can be used as a windscreen for the microphone.



A small amount of foam can be used as a windshield for the microphone.

To prevent insects from entering the device, a piece of fine mesh can be installed over the opening with MicrophoneProtector.stl.

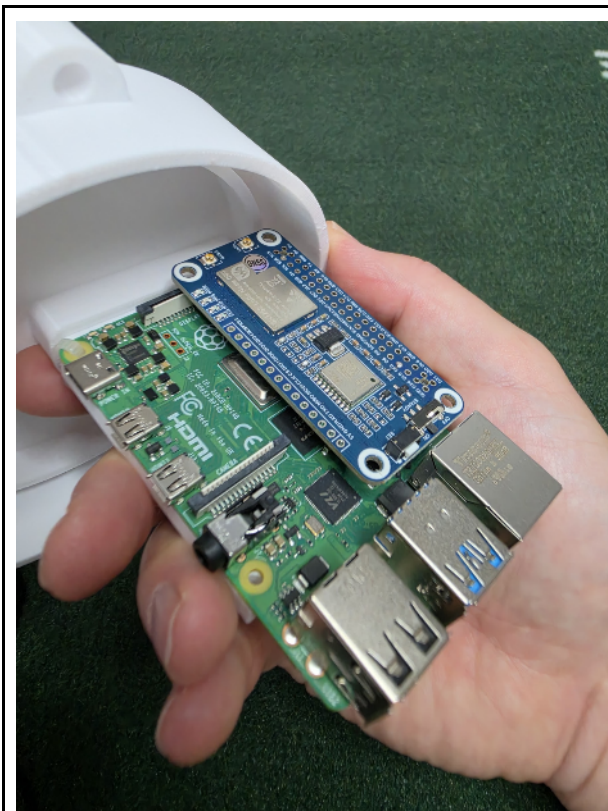
Existing solutions can also be used: one half of a tea strainer provides a convenient and inexpensive insect screen.



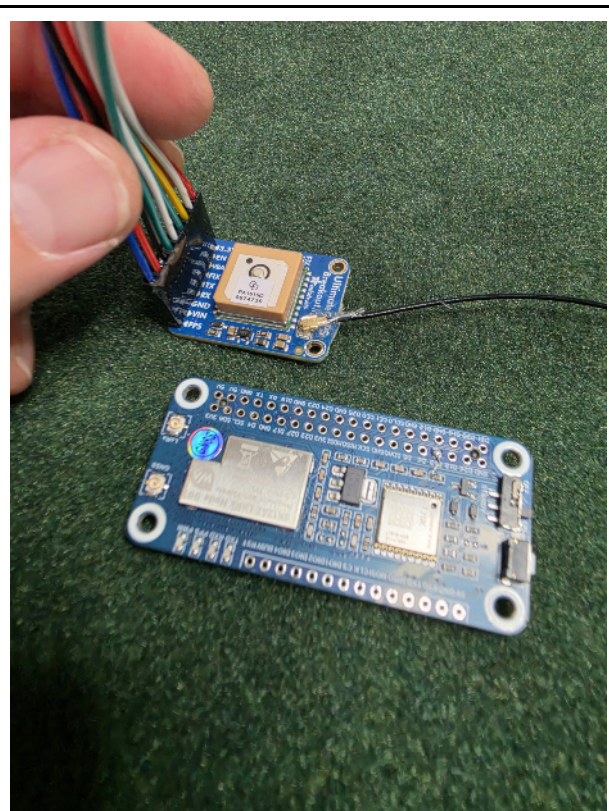
Sensitive microphones can be affected by electromagnetic interference and acoustic resonance inside the enclosure. To reduce unwanted noise, the microphone should be isolated from vibrating surfaces, and cable routing should avoid unnecessary contact with the microphone body. The Dodotronic microphone mount provides a tight fit that minimizes acoustic resonance. If an optional LTE modem is used, it is recommended to place it outside the enclosure and as far from the microphone as practical to reduce potential radio-frequency interference.

GNSS

A GNSS receiver can be added to the device to provide location data and an accurate time reference when no internet connection is available. Several GNSS modules are available for Raspberry Pi, including HAT expansion boards and compact receiver options. In this prototype, a Waveshare SX1262 module is used. Adding GNSS improves the timing accuracy of audio recordings and helps locate deployed devices in the field.



Waveshare SX1262



Waveshare SX1262 and small GNSS alternative with integrated ceramic antenna

External antennas can be mounted at a suitable distance from the device using a structure like the one shown below. Cable pass-through holes for the antenna connections should be drilled in the rear end cap of the enclosure. Positioning antennas outside the enclosure and away from sensitive electronics can help improve signal quality and reduce interference.



The following files are needed for antenna assembly:

- external-antenna-structure-LTE-holder.stl
- external-antenna-structure-Plate.stl
- external-antenna-structure-Tree-Attachment.stl

WATERPROOFING CONSIDERATIONS

The device is not completely waterproof. The following measures are recommended to improve resistance to rain, splashes, and moisture, but they do not make the enclosure suitable for submersion or continuous exposure to standing water:

- Seal all external cable entries.
- Silicone sealant can be applied to the groove of the end caps; however, the pipe must be cut perfectly to ensure a proper seal.
- Additional waterproof tape can be applied on top of the device's assembly joints for extra protection.
- If there is no need to open the front-end cap during operation, it can be permanently glued in place. In that case, only the rear end cap should remain removable for maintenance and access.

MOUNTING AND SECURITY



The device can be mounted using the holes integrated into the end caps and a rubber-coated steel cable, for example to a tree, pole, or similar structure. The cable can be locked to reduce the risk of unauthorized opening, removal, or component theft. The mounting point should be stable enough to prevent movement that could create unwanted noise in the recordings.